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Production-Grade Movie Recommendation System

Project Overview

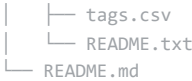
This project implements an industry-standard movie recommendation engine designed to enhance user engagement and retention on digital entertainment platforms. By leveraging advanced supervised modeling techniques, including Collaborative Filtering (SVD), Content-Based Filtering, and Neural Collaborative Filtering (NCF), the system provides highly personalized movie suggestions that address the "choice paralysis" and "cold start" challenges common in the streaming industry.

Key Features

- **Hybrid Recommendation Engine:** Combines the behavioral insights of Collaborative Filtering with the attribute-based precision of Content-Based Filtering.
- **Advanced Evaluation Metrics:** Utilizes NDCG@10 (Normalized Discounted Cumulative Gain) and Precision@K to measure ranking effectiveness, moving beyond simple rating accuracy (RMSE).
- **Temporal Validation Strategy:** Implements a time-based train-test split to ensure model performance on future, unseen data, simulating a real-world production environment.
- **Deep Learning Baseline:** Introduces Neural Collaborative Filtering (NCF) using PyTorch to capture complex, non-linear user-item interactions.
- **Cold Start Solution:** Employs movie metadata (genres and tags) to provide intelligent recommendations for new content and users with sparse history.

Repository Structure

```
.
├── Industry_Standard_Recommendation_System.ipynb # Main project notebook with analysis and modeling
├── ml-latest-small/                               # Dataset directory (MovieLens 100k)
│   ├── links.csv
│   ├── movies.csv
│   └── ratings.csv
```



Project documentation

Getting Started

Prerequisites

Ensure you have the following Python packages installed:

- pandas
- numpy
- scikit-learn
- scikit-surprise
- torch

Installation

1. Clone the repository:

```
git clone https://github.com/your-username/Movie-Recommendation-System.git
cd Movie-Recommendation-System
```



2. Install dependencies:

```
pip install pandas numpy scikit-learn scikit-surprise torch
```



Running the Project

Open the `Industry_Standard_Recommendation_System.ipynb` notebook in a Jupyter environment to view the full analysis, modeling process, and evaluation results.

Data Source

The project uses the [MovieLens ml-latest-small dataset](#), which contains 100,836 ratings and 3,683 tag applications across 9,742 movies by 610 users.

Evaluation Results

Metric	SVD Model Performance
RMSE	0.8775
NDCG@10	0.8558

Business Impact

- **Increased Engagement:** Personalized discovery leads to longer user sessions.
- **Reduced Churn:** Relevant recommendations improve subscriber satisfaction and retention.
- **Content ROI:** Ensures new and niche content is discoverable, maximizing the value of the entire library.

Future Roadmap

- **Real-time Optimization:** Implementing streaming data pipelines for instant profile updates.
- **Model Explainability:** Integrating tools like LIME to provide transparent "Why this was recommended" features.

- Enterprise Scalability: Transitioning to cloud-based Spark environments for global-scale deployment.

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

Contributors 1



Gesare-n Nicole Gesare Onyiego

Languages

● Jupyter Notebook 100.0%